

BEYOND THE SPECTRUM: FUNCTIONAL REALIZATIONS OF MATRICES AND TENSORS, EMERGING INVARIANTS, AND GEOMETRIC MEASURES

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ABSTRACT. Finite-dimensional multilinear algebra treats matrices and higher-order tensors as elements of discrete linear spaces, extracting intrinsic invariants such as eigenvalues, singular values, operator rank, and trace polynomials. However, pure algebraic representations fundamentally obscure spatial locality, topological connectivity, and geometric regularity due to basis-permutation invariance in coordinates. In this treatise, we introduce a unified mathematical framework termed *Functional Realization Operators* $\Phi: \mathcal{T}^k(V) \rightarrow \mathcal{F}(\Omega)$, mapping discrete algebraic arrays into function spaces over continuous domains Ω . We construct an exhaustive taxonomy of continuous, piecewise continuous, step-like, and operator-valued realization families.

We rigorously prove that embedding matrices and tensors into function spaces unleashes an entire hierarchy of *emerging invariants* completely unreachable by pure matrix algebra alone: (i) Dirichlet energies and Sobolev $H^s(\Omega)$ regularities that break spectral permutation symmetry; (ii) bounded variation (BV) norms and geometric Hausdorff measures of level sets via the Coarea formula; (iii) Morse-theoretic critical point indices, handlebody decompositions, and Euler–Poincaré topological characteristics on spheres; and (iv) the cut-norm topology of graphons capturing non-trivial continuum limits as the ambient matrix dimensions diverge to infinity ($n \rightarrow \infty$).

Crucially, we demonstrate five paradigm-shifting implications of this framework across modern science: (1) breaking spectral blindness in Deep Learning via Sobolev weight regularizations and infinite-width graphon scaling; (2) unlocking spin glass free-energy landscapes in statistical physics through spherical Morse topology; (3) defining universal, scale-free metrics on massive growing networks; (4) circumventing the non-closedness pathologies of De Silva–Lim in computational tensor rank approximation; and (5) providing a geometric Hausdorff perimeter foundation for medical imaging and minimal surface theory.

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1. INTRODUCTION AND MOTIVATION

Classical linear and multilinear algebra views a real matrix $A \in \mathbb{R}^{m \times n}$ or a tensor $\mathcal{T} \in \mathbb{R}^{d_1 \times \dots \times d_k}$ through the lens of coordinate arrays and linear maps between finite-dimensional vector spaces. Within this paradigm, the dominant structural invariants are purely algebraic: the trace $\text{Tr}(A)$, the determinant $\det(A)$, the characteristic polynomial $P_A(\lambda)$, the spectrum $\sigma(A) = \{\lambda_i(A)\}$, the singular value spectrum $\Sigma(A) = \{\sigma_i(A)\}$, and the matrix rank $\text{rank}(A)$.

While undeniably foundational, this purely algebraic framework suffers from a significant limitation when matrices represent spatial, physical, network, or image-based structures: *algebraic invariants are oblivious to metric and topological spatial layouts*. Specifically:

- (a) **Permutation Invariance vs. Metric Proximity:** If $P \in \mathbb{R}^{n \times n}$ is a permutation matrix, the conjugated matrix $B = PAP^T$ possesses an identical spectrum, trace, determinant, and Frobenius norm:

$$\sigma(PAP^T) = \sigma(A), \quad \text{Tr}(PAP^T) = \text{Tr}(A), \quad \|PAP^T\|_F = \|A\|_F. \quad (1.1)$$

Yet, in physical space, transposing two adjacent rows versus swapping two distant rows induces fundamentally different topological discontinuities, gradient concentrations, and oscillatory phenomena.

- (b) **The Curse of Dimensional Rigidity** ($n \rightarrow \infty$): Two matrices of differing sizes, say $A_{100 \times 100}$ and $B_{1000 \times 1000}$, live in non-isomorphic vector spaces. Algebraic linear algebra provides no intrinsic metric to measure how closely A and B approximate the same underlying continuum limit object without ad-hoc downsampling or artificial padding.
- (c) **Higher-Order Tensor Intractability:** For tensors $\mathcal{T} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ of order $k \geq 3$, algebraic invariants become notoriously pathological: tensor rank determination is NP-hard over $\mathbb{R}$ and $\mathbb{C}$, low-rank approximations can fail to exist due to non-closed orbits (border rank pathologies), and multilinear spectral theory yields complicated varieties rather than discrete eigenspaces.

To transcend these limitations, one must bridge the gap between multilinear algebra and *infinite-dimensional functional analysis, geometric measure theory, and differential topology*.

Core Thesis and Contributions. This paper demonstrates that by formally mapping matrices and tensors into function spaces via *Functional Realization Operators* $\Phi: \mathcal{T}^k(V) \rightarrow \mathcal{F}(\Omega)$, we unlock a spectrum of geometric and analytical invariants that:

1. Discriminate between spectrally indistinguishable arrays according to spatial energy, harmonic smoothness, and Sobolev scale $H^s(\Omega)$;
2. Measure internal perimeter and level-set geometry through functions of bounded variation (BV) and the geometric Coarea formula;
3. Unveil the non-convex optimization topology through Morse critical point indices on compact manifolds $(\mathbb{S}^{n-1})$;
4. Establish continuous limits of dense discrete arrays via the cut-norm topology and graphons ($L^\infty \rightarrow L^1$ operator duality);
5. Provide rigorous foundational solutions to five major open challenges across machine learning, spin glass physics, large networks, tensor approximation, and medical imaging.

2. AXIOMATIC FRAMEWORK: FUNCTIONAL REALIZATION OPERATORS

Let $V_1, \dots, V_k$ be real vector spaces of finite dimensions $d_1, \dots, d_k$ equipped with standard inner products $\langle \cdot, \cdot \rangle_{V_i}$, and let $\mathcal{T}^k(V_1, \dots, V_k) \cong V_1 \otimes \dots \otimes V_k$ denote the k -th order tensor product space. When $V_i = \mathbb{R}^{d_i}$ with canonical bases $\{e_j^{(i)}\}$, any tensor $\mathcal{T} \in \mathcal{T}^k(V_1, \dots, V_k)$ is uniquely identified with its coordinate multi-array $(T_{j_1 \dots j_k})_{1 \leq j_r \leq d_r}$. For $k = 2$, this reduces to the classical matrix space $\mathbb{R}^{d_1 \times d_2}$.

Let $\Omega \subseteq \mathbb{R}^d$ be a measurable domain endowed with the Lebesgue measure dx , or a smooth compact Riemannian manifold (M, g) with volume form dV_g . We denote by $\mathcal{F}(\Omega)$ a topological vector space of functions over Ω (e.g., $C^0(\Omega)$, $L^p(\Omega)$, $W^{s,p}(\Omega)$, or the space of Radon measures $\mathcal{M}(\Omega)$).

Definition 2.1 (Functional Realization Operator). A *Functional Realization Operator* is a mapping

$$\Phi: \mathcal{T}^k(V_1, \dots, V_k) \longrightarrow \mathcal{F}(\Omega) \quad (2.1)$$

which associates to each discrete tensor $\mathcal{T}$ a well-defined function $f_{\mathcal{T}} := \Phi(\mathcal{T}) \in \mathcal{F}(\Omega)$.

Definition 2.2 (Structural Properties of Realizations). Let $\Phi: \mathcal{T}^k \rightarrow \mathcal{F}(\Omega)$ be a functional realization. We categorize Φ according to the following properties:

- (i) **Linearity:** Φ is linear if $\Phi(\alpha\mathcal{S} + \beta\mathcal{T}) = \alpha\Phi(\mathcal{S}) + \beta\Phi(\mathcal{T})$ for all $\alpha, \beta \in \mathbb{R}$ and $\mathcal{S}, \mathcal{T} \in \mathcal{T}^k$.
- (ii) **Injectivity (Faithfulness):** Φ is faithful if $\ker(\Phi) = \{0\}$, meaning no non-zero tensor vanishes identically as a continuous function.
- (iii) **Continuity (Boundedness):** If $\mathcal{T}^k$ is equipped with the Frobenius norm $\|\mathcal{T}\|_F = \sqrt{\sum T_{j_1 \dots j_k}^2}$ and $\mathcal{F}(\Omega)$ with norm $\|\cdot\|_{\mathcal{F}}$, Φ is bounded if there exists $C > 0$ such that

$$\|\Phi(\mathcal{T})\|_{\mathcal{F}} \leq C\|\mathcal{T}\|_F, \quad \forall \mathcal{T} \in \mathcal{T}^k. \quad (2.2)$$

- (iv) **Symmetry Equivariance:** Let G be a subgroup of the product of orthogonal groups $O(d_1) \times \dots \times O(d_k)$ (or permutation groups $\mathcal{S}_{d_1} \times \dots \times \mathcal{S}_{d_k}$) acting on $\mathcal{T}^k$, and let G act on Ω via measurable transformations $\tau_g: \Omega \rightarrow \Omega$. Φ is G -equivariant if

$$\Phi(g \cdot \mathcal{T})(x) = \Phi(\mathcal{T})(\tau_{g^{-1}}(x)), \quad \forall x \in \Omega, \forall g \in G. \quad (2.3)$$

3. COMPREHENSIVE TAXONOMY OF FUNCTIONAL REALIZATIONS

3.1. Family I: Multilinear, Polynomial, and Spherical Realizations. In this family, the matrix or tensor coefficients define the homogeneous algebraic form of a polynomial on Euclidean space or a compact sphere.

Construction 3.1 (Quadratic Forms on $\mathbb{R}^n$ and $\mathbb{S}^{n-1}$). Let $A \in \mathbb{R}^{n \times n}$. The quadratic realization $\Phi_{\text{quad}}(A): \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by:

$$f_A(x) := x^T A x = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j. \quad (3.1)$$

Restricting x to the unit sphere $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n: \|x\|_2 = 1\}$ yields a smooth function $f_A \in C^\infty(\mathbb{S}^{n-1})$.

Construction 3.2 (Higher-Order Homogeneous Tensor Forms). For a d -th order symmetric tensor $\mathcal{T} \in \text{Sym}^d(\mathbb{R}^n)$, the homogeneous realization $\Phi_{\text{hom}}(\mathcal{T}): \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ is:

$$f_{\mathcal{T}}(x) := \mathcal{T}(x, x, \dots, x) = \sum_{i_1, \dots, i_d=1}^n \mathcal{T}_{i_1 \dots i_d} x_{i_1} \dots x_{i_d}. \quad (3.2)$$

Proposition 3.3 (Spectral and Critical Correspondence). *Let $A \in \text{Sym}_n(\mathbb{R})$ be symmetric with eigendecomposition $A = V\Lambda V^T$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. Then:*

- (i) *The global extrema of $f_A|_{\mathbb{S}^{n-1}}$ satisfy:*

$$\min_{x \in \mathbb{S}^{n-1}} f_A(x) = \lambda_1, \quad \max_{x \in \mathbb{S}^{n-1}} f_A(x) = \lambda_n. \quad (3.3)$$

- (ii) *The critical points of the function $f_A: \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ are precisely the normalized eigenvectors v_i of A , and the critical values are the corresponding eigenvalues λ_i .*

Proof. Consider the Lagrangian $\mathcal{L}(x, \mu) = x^T A x - \mu(x^T x - 1)$ on $\mathbb{R}^n \times \mathbb{R}$. The Euler-Lagrange criticality condition $\nabla_x \mathcal{L} = 2Ax - 2\mu x = 0$ immediately yields the eigenvalue equation $Ax = \mu x$. Constraining $\|x\|_2 = 1$ gives $f_A(x) = x^T A x = \mu x^T x = \mu$. Thus, the critical values are precisely the spectrum $\sigma(A)$, and the extrema coincide with the Rayleigh quotient bounds. $\square$

3.2. Family II: Step-like, Partition-Based, and Graphon Realizations. Here, the discrete array parametrizes a piecewise-constant function over a partitioned measure space. This construction forms the foundation of modern dense graph limit theory (Graphons) and digital imaging.

Construction 3.4 (Graphon / Unit Square Step Realization). Let $A \in \mathbb{R}^{n \times n}$ and consider the unit square $\Omega = [0, 1]^2$ with Lebesgue measure. Partition $[0, 1]$ into n equal left-closed, right-open intervals:

$$I_i := \left[\frac{i-1}{n}, \frac{i}{n} \right), \quad i = 1, \dots, n-1, \quad \text{and} \quad I_n := \left[\frac{n-1}{n}, 1 \right]. \quad (3.4)$$

The graphon step realization $\Phi_{\text{step}}(A): [0, 1]^2 \rightarrow \mathbb{R}$ is defined by:

$$W_A(x, y) := \sum_{i=1}^n \sum_{j=1}^n a_{ij} \mathbf{1}_{I_i}(x) \mathbf{1}_{I_j}(y), \quad (3.5)$$

where $\mathbf{1}_E$ denotes the characteristic function of set E .

Construction 3.5 (Multidimensional Pixel/Voxel Fields). Let $A \in \mathbb{R}^{n_1 \times \dots \times n_d}$ be a d -dimensional tensor. Partition $\Omega = \prod_{\alpha=1}^d [0, L_\alpha]$ into hypercubes $Q_{\mathbf{i}} = \prod_{\alpha=1}^d [(i_\alpha - 1)h_\alpha, i_\alpha h_\alpha]$. The realization is:

$$f_A(x) = \sum_{\mathbf{i}} A_{\mathbf{i}} \mathbf{1}_{Q_{\mathbf{i}}}(x). \quad (3.6)$$

Notice that $f_A \notin C^0(\Omega)$; rather, $f_A \in L^\infty(\Omega) \cap \text{BV}(\Omega)$.

Construction 3.6 (Polyhedral Indicator Functions and Hyperplane Arrangements). Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. The matrix A defines a system of m affine half-spaces $H_i^- = \{x \in \mathbb{R}^n : \langle a_i, x \rangle \leq b_i\}$, where a_i is the i -th row of A . The polyhedral indicator realization is:

$$\Phi_{\text{poly}}(A, b)(x) := \mathbf{1}_{\{x \in \mathbb{R}^n : Ax \leq b\}}(x) = \prod_{i=1}^m \Theta(b_i - \langle a_i, x \rangle), \quad (3.7)$$

where $\Theta(t) = \mathbf{1}_{[0, \infty)}(t)$ is the Heaviside step function.

3.3. Family III: Interpolation, Basis Expansions, and Sobolev Fields. In this family, matrix elements act as generalized Fourier coefficients, spline control points, or localized radial basis weights.

Construction 3.7 (Harmonic / Dual Fourier Realization). Let $A \in \mathbb{C}^{n \times n}$. Let $\mathbb{T}^2 = [0, 2\pi)^2$ be the 2-torus equipped with the normalized Haar probability measure $d\mu(x) = \frac{dx_1 dx_2}{(2\pi)^2}$. We associate A with the trigonometric polynomial:

$$f_A(x_1, x_2) := \sum_{k_1, k_2=1}^n a_{k_1 k_2} e^{i(k_1 x_1 + k_2 x_2)}. \quad (3.8)$$

By Parseval's identity on $\{e^{i(k_1 x_1 + k_2 x_2)}\}$, the $L^2(\mathbb{T}^2, d\mu)$ norm satisfies $\|f_A\|_{L^2(\mathbb{T}^2)} = \|A\|_F$.

Construction 3.8 (Tensor Product Splines and Bézier Surfaces). Let $B_{i,p}(u)$ denote the i -th B-spline basis function of degree p on a knot vector Ξ . For a matrix of control points $P \in \mathbb{R}^{m \times n \times 3}$:

$$S(u, v) = \sum_{i=0}^m \sum_{j=0}^n B_{i,p}(u) B_{j,q}(v) P_{ij}. \quad (3.9)$$

Here, the matrix values control continuity (C^{p-1} regularity), convex hull containment, and geometric curvature forms.

3.4. Family IV: Integral Operators and Continuous Matrix Limits.

Construction 3.9 (Fredholm Integral Kernel Operators). Every matrix $A \in \mathbb{R}^{n \times n}$ can be identified with the integral operator $T_A: L^2([0, 1]) \rightarrow L^2([0, 1])$:

$$(T_A u)(x) := \int_0^1 W_A(x, y) u(y) dy, \quad (3.10)$$

where W_A is the step realization given in (3.5). T_A is a Hilbert-Schmidt operator with Hilbert-Schmidt norm $\|T_A\|_{\text{HS}} = \frac{1}{n} \|A\|_F$.

4. EMERGING INVARIANTS: WHAT MATRIX ALGEBRA CANNOT SEE

4.1. Spatiality vs. Spectrum: Dirichlet Energies and Sobolev Regularity. Let $A \in \mathbb{R}^{n \times n}$. In linear algebra, the permutation matrix P_π corresponding to a permutation $\pi \in \mathcal{S}_n$ preserves all spectral invariants:

$$\sigma(P_\pi A P_\pi^T) = \sigma(A), \quad \text{Tr}(P_\pi A P_\pi^T) = \text{Tr}(A), \quad \|P_\pi A P_\pi^T\|_F = \|A\|_F. \quad (4.1)$$

Now consider the harmonic realization on $(\mathbb{T}^2, d\mu)$ given by (3.8):

$$f_A(x_1, x_2) = \sum_{k_1, k_2=1}^n a_{k_1 k_2} e^{i(k_1 x_1 + k_2 x_2)}. \quad (4.2)$$

The homogeneous Dirichlet energy under normalized measure $d\mu(x) = \frac{dx_1 dx_2}{(2\pi)^2}$ is:

$$\mathcal{E}(f) := \|\nabla f\|_{L^2(\mathbb{T}^2, d\mu)}^2 = \int_{\mathbb{T}^2} \|\nabla f(x_1, x_2)\|_2^2 d\mu(x). \quad (4.3)$$

Theorem 4.1 (Spectral Blindness to Spatial Energy). *For any matrix $A \in \mathbb{R}^{n \times n}$, the Dirichlet energy of the realization f_A is given explicitly by:*

$$\mathcal{E}(f_A) = \sum_{k_1=1}^n \sum_{k_2=1}^n (k_1^2 + k_2^2) |a_{k_1 k_2}|^2. \quad (4.4)$$

Moreover, there exist non-zero symmetric matrices $A \in \text{Sym}_n(\mathbb{R})$ and orthogonal permutation matrices $P_1, P_2 \in \mathcal{S}_n$ such that $A_1 = P_1 A P_1^T$ and $A_2 = P_2 A P_2^T$ satisfy:

$$\sigma(A_1) = \sigma(A_2) \quad \text{and} \quad \|A_1\|_F = \|A_2\|_F = \|A\|_F, \quad (4.5)$$

yet the ratio of their functional Dirichlet energies scales as:

$$\frac{\mathcal{E}(f_{A_2})}{\mathcal{E}(f_{A_1})} = \Theta(n^2). \quad (4.6)$$

Remark 4.2 (Isotropic Invariance vs Anisotropic Discrepancy). For isotropic matrices such as $A = cI_n$ (scalar multiples of the identity) or $A = cJ_n$ (the all-ones matrix), any permutation matrix satisfies $PAP^T = A$, yielding an energy ratio $\mathcal{E}(f_{A_2})/\mathcal{E}(f_{A_1}) = 1$. The maximal quadratic amplification $\Theta(n^2)$ is achieved by anisotropic and localized configurations, such as projectors and rank-one forms.

Proof. Direct differentiation of f_A yields:

$$\partial_{x_1} f_A = \sum_{k_1, k_2=1}^n (ik_1) a_{k_1 k_2} e^{i(k_1 x_1 + k_2 x_2)}, \quad \partial_{x_2} f_A = \sum_{k_1, k_2=1}^n (ik_2) a_{k_1 k_2} e^{i(k_1 x_1 + k_2 x_2)}. \quad (4.7)$$

Since this is a finite trigonometric polynomial, termwise differentiation and the interchange of limits are unconditionally justified. By Parseval's identity on the orthogonal character system $\{e^{i(k_1 x_1 + k_2 x_2)}\}$:

$$\|\nabla f_A\|_{L^2}^2 = \|\partial_{x_1} f_A\|_{L^2}^2 + \|\partial_{x_2} f_A\|_{L^2}^2 = \sum_{k_1, k_2=1}^n (k_1^2 + k_2^2) |a_{k_1 k_2}|^2. \quad (4.8)$$

To prove the ratio bound, consider a rank-1 matrix $A = uu^T$ where $u \in \mathbb{R}^n$ has a single non-zero component $u_1 = 1$. Let $\pi_1 = \text{id}$ so that $A_1 = E_{1,1}$, which places the entire spectral mass at the base frequency index $(1, 1)$. Then:

$$\mathcal{E}(f_{A_1}) = (1^2 + 1^2) \cdot 1^2 = 2. \quad (4.9)$$

Now let π_2 swap index 1 and n , so $A_2 = P_{\pi_2} A P_{\pi_2}^T = E_{n,n}$, concentrating the spectral mass at maximal frequency (n, n) . Then:

$$\mathcal{E}(f_{A_2}) = (n^2 + n^2) \cdot 1^2 = 2n^2. \quad (4.10)$$

Both A_1 and A_2 possess an identical spectrum $\sigma = \{1, 0, \dots, 0\}$, identical rank 1, and identical Frobenius norm $\|A_1\|_F = \|A_2\|_F = 1$. However:

$$\frac{\mathcal{E}(f_{A_2})}{\mathcal{E}(f_{A_1})} = \frac{2n^2}{2} = n^2 = \Theta(n^2). \quad (4.11)$$

This completes the proof. $\square$

Example 4.3 (The Checkerboard Matrix Pathology). Consider the checkerboard array $A \in \mathbb{R}^{n \times n}$ given by $a_{ij} = (-1)^{i+j}$. In discrete linear algebra, $A = vv^T$ with $v = (1, -1, 1, -1, \dots)^T$ is an elementary rank-1 matrix with spectrum $\sigma(A) = \{n, 0, \dots, 0\}$. Pure algebra classifies A as structurally simple. However, under the harmonic realization, every entry has $|a_{k_1 k_2}|^2 = 1$, yielding by (4.4) an explosive Dirichlet energy:

$$\mathcal{E}(f_A) = \sum_{k_1=1}^n \sum_{k_2=1}^n (k_1^2 + k_2^2) \cdot 1 = 2n \sum_{k=1}^n k^2 = 2n \cdot \frac{n(n+1)(2n+1)}{6} = \frac{2}{3}n^4 + \mathcal{O}(n^3) = \Theta(n^4). \quad (4.12)$$

Functionally, the matrix represents the maximally rough spatial signal.

4.2. Geometric Measure Theory: Bounded Variation and the Coarea Formula. We now investigate the step realization $\Phi_{\text{step}}(A)$ on $\Omega = (0, 1)^2$. Because W_A is discontinuous across grid edges, its classical gradient does not exist. However, W_A belongs to the space of *Functions of Bounded Variation* $\text{BV}(\Omega)$.

Definition 4.4 (Total Variation in $\text{BV}(\Omega)$). Let $\Omega \subset \mathbb{R}^2$ be an open domain. For $f \in L^1(\Omega)$, the *Total Variation* of f is defined duality-wise by:

$$\text{TV}(f) := \sup \left\{ \int_{\Omega} f(x) \operatorname{div} \varphi(x) \, dx : \varphi \in C_c^1(\Omega; \mathbb{R}^2), \|\varphi\|_{L^\infty} \leq 1 \right\}. \quad (4.13)$$

Theorem 4.5 (Total Variation of the Matrix Step Realization). *Let $A \in \mathbb{R}^{n \times n}$ and let $W_A \in \text{BV}((0, 1)^2)$ be its step realization defined on the open unit square $(0, 1)^2$. Then the distributional gradient DW_A is a 1-dimensional Radon measure concentrated on the grid lines $\mathcal{G}_n = \bigcup_{i=1}^{n-1} (\{i/n\} \times [0, 1]) \cup \bigcup_{j=1}^{n-1} ([0, 1] \times \{j/n\})$, and its total variation is given by:*

$$\text{TV}(W_A) = \frac{1}{n} \sum_{i=1}^{n-1} \sum_{j=1}^n |a_{i+1,j} - a_{i,j}| + \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^{n-1} |a_{i,j+1} - a_{i,j}|. \quad (4.14)$$

Proof. Let $\varphi = (\varphi_1, \varphi_2) \in C_c^1((0, 1)^2; \mathbb{R}^2)$ with $\|\varphi\|_{L^\infty} \leq 1$. By Green-Gauss integration by parts over each sub-square $Q_{ij} = (\frac{i-1}{n}, \frac{i}{n}) \times (\frac{j-1}{n}, \frac{j}{n})$:

$$\int_{(0,1)^2} W_A \operatorname{div} \varphi \, dx \, dy = \sum_{i,j=1}^n a_{ij} \int_{Q_{ij}} \left(\frac{\partial \varphi_1}{\partial x} + \frac{\partial \varphi_2}{\partial y} \right) \, dx \, dy. \quad (4.15)$$

Applying the divergence theorem on each cell gives:

$$\int_{Q_{ij}} \operatorname{div} \varphi \, dx \, dy = \int_{\partial Q_{ij}} \varphi \cdot \nu \, d\mathcal{H}^1, \quad (4.16)$$

where ν is the outward unit normal and $\mathcal{H}^1$ is the 1-dimensional Hausdorff measure.

At the internal vertical boundary separating Q_{ij} and $Q_{i+1,j}$, denoted $\Gamma_{i,j}^v = \{i/n\} \times (\frac{j-1}{n}, \frac{j}{n})$, the outward normal from Q_{ij} is $(1, 0)$ and from $Q_{i+1,j}$ is $(-1, 0)$. Summing across all interfaces:

$$\begin{aligned} \int_{(0,1)^2} W_A \operatorname{div} \varphi \, dx \, dy &= \sum_{i=1}^{n-1} \sum_{j=1}^n (a_{ij} - a_{i+1,j}) \int_{\Gamma_{i,j}^v} \varphi_1 \left(\frac{i}{n}, y \right) \, dy \\ &\quad + \sum_{i=1}^n \sum_{j=1}^{n-1} (a_{ij} - a_{i,j+1}) \int_{\Gamma_{i,j}^h} \varphi_2 \left(x, \frac{j}{n} \right) \, dx. \end{aligned} \quad (4.17)$$

Taking the supremum over test vector fields with support concentrated in disjoint tubular neighborhoods of the interface segments allows us to saturate the sign of each difference independently on each segment of length $1/n$:

$$\sup_{\|\varphi\| \leq 1} \int_{(0,1)^2} W_A \operatorname{div} \varphi \, dx \, dy = \sum_{i=1}^{n-1} \sum_{j=1}^n |a_{i+1,j} - a_{i,j}| \cdot \frac{1}{n} + \sum_{i=1}^n \sum_{j=1}^{n-1} |a_{i,j+1} - a_{i,j}| \cdot \frac{1}{n}. \quad (4.18)$$

This establishes (4.14). $\square$

Theorem 4.6 (Geometric Coarea Linkage: Hausdorff Measure of Matrix Level Curves). *For any $t \in \mathbb{R}$, define the continuous sub-level set $E_t := \{(x, y) \in (0, 1)^2 : W_A(x, y) > t\}$. Then E_t is a set of finite perimeter in $(0, 1)^2$, and its perimeter $\operatorname{Per}(E_t; (0, 1)^2) = \mathcal{H}^1(\partial^* E_t)$ satisfies the Coarea Formula:*

$$\operatorname{TV}(W_A) = \int_{-\infty}^{\infty} \mathcal{H}^1(\partial^* E_t) \, dt, \quad (4.19)$$

where $\partial^* E_t$ is the essential boundary of E_t in the sense of De Giorgi.

Proof. Since $W_A \in \operatorname{BV}((0, 1)^2)$ by Theorem 4.5, De Giorgi's structure theorem guarantees that for almost every $t \in \mathbb{R}$, E_t is a Caccioppoli set of finite perimeter and the classical Federer Coarea Identity holds (see Federer [6], Ambrosio, Fusco, and Pallara [2], and Evans and Gariepy [5]). $\square$

4.3. Differential Topology: Morse Theory and Landscape Homology. We now examine the polynomial realization on the sphere $\mathbb{S}^{n-1}$, defined in Section 3.1:

$$f_A(x) = x^T A x, \quad x \in \mathbb{S}^{n-1}. \quad (4.20)$$

Assume that $A \in \operatorname{Sym}_n(\mathbb{R})$ is symmetric.

Theorem 4.7 (Morse Spectrum of Quadratic Realizations). *Let $A \in \operatorname{Sym}_n(\mathbb{R})$ have strictly distinct eigenvalues:*

$$\lambda_1 < \lambda_2 < \cdots < \lambda_n. \quad (4.21)$$

Then $f_A: \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ is a strictly non-degenerate Morse function. Moreover:

(i) f_A has exactly $2n$ isolated critical points, occurring in antipodal pairs:

$$\operatorname{Crit}(f_A) = \{\pm v_1, \pm v_2, \dots, \pm v_n\}, \quad (4.22)$$

where v_i is the normalized eigenvector corresponding to λ_i .

(ii) The Morse index of the critical points $\pm v_i$ is given precisely by:

$$\gamma(\pm v_i) = i - 1. \quad (4.23)$$

(iii) The Morse polynomial $P_t(f_A) := \sum_{p \in \text{Crit}(f_A)} t^{\gamma(p)}$ satisfies:

$$P_t(f_A) = 2 \sum_{i=1}^n t^{i-1} = 2(1 + t + t^2 + \cdots + t^{n-1}). \quad (4.24)$$

Evaluating at $t = -1$ rigorously recovers the topological Euler–Poincaré characteristic:

$$\chi(\mathbb{S}^{n-1}) = P_{-1}(f_A) = 1 - (-1)^n = \begin{cases} 0, & n \text{ even } (\mathbb{S}^{n-1} \text{ odd}), \\ 2, & n \text{ odd } (\mathbb{S}^{n-1} \text{ even}). \end{cases} \quad (4.25)$$

Proof. Let $x \in \mathbb{S}^{n-1}$. The Riemannian gradient on the sphere with respect to the standard round metric is the projection of the ambient gradient onto $T_x \mathbb{S}^{n-1} = \{u \in \mathbb{R}^n : \langle u, x \rangle = 0\}$:

$$\nabla_{\mathbb{S}^{n-1}} f_A(x) = 2Ax - 2(x^T Ax)x. \quad (4.26)$$

Thus, $\nabla_{\mathbb{S}^{n-1}} f_A(x) = 0$ if and only if $Ax = (x^T Ax)x$. Since $\|x\|_2 = 1$, x is an eigenvector with eigenvalue $\lambda = x^T Ax$. Because the eigenvalues are strictly distinct, the only unit critical points are $x = \pm v_i$ for $i = 1, \dots, n$.

Next, we evaluate the Riemannian Hessian at $x = v_i$. Along a geodesic $\gamma(t)$ on $\mathbb{S}^{n-1}$ with $\gamma(0) = v_i$ and $\dot{\gamma}(0) = u \in T_{v_i} \mathbb{S}^{n-1}$, the geodesic curvature satisfies $\ddot{\gamma}(0) = -\|u\|_2^2 v_i$. Differentiating:

$$\text{Hess}_{\mathbb{S}^{n-1}} f_A(v_i)(u, u) = \left. \frac{d^2}{dt^2} f_A(\gamma(t)) \right|_{t=0} = 2u^T Au - 2\lambda_i \|u\|_2^2. \quad (4.27)$$

Expanding u in the orthonormal eigenbasis $\{v_j\}_{j \neq i}$: $u = \sum_{j \neq i} c_j v_j$. Then:

$$\text{Hess}_{\mathbb{S}^{n-1}} f_A(v_i)(u, u) = 2 \sum_{j \neq i} (\lambda_j - \lambda_i) c_j^2. \quad (4.28)$$

Since $\lambda_1 < \cdots < \lambda_n$:

- $(\lambda_j - \lambda_i) < 0$ for $j \in \{1, \dots, i-1\}$, producing exactly $i-1$ negative eigenvalues;
- $(\lambda_j - \lambda_i) > 0$ for $j \in \{i+1, \dots, n\}$, producing $(n-1) - (i-1) = n-i$ positive eigenvalues;
- there are no zero eigenvalues since $\lambda_j \neq \lambda_i$ for all $j \neq i$.

Thus, $\ker(\text{Hess}_{\mathbb{S}^{n-1}} f_A(v_i)) = \{0\}$ and the Morse index is $\gamma(\pm v_i) = i - 1$.

The Morse polynomial evaluates to $P_{-1}(f_A) = 2 \sum_{i=1}^n (-1)^{i-1} = 2 \cdot \frac{1 - (-1)^n}{2} = 1 - (-1)^n$. By classical Morse theory (see Milnor [11]), this alternating sum of Morse indices exactly recovers the Euler characteristic $\chi(\mathbb{S}^{n-1})$ obtained from singular homology $H_*(\mathbb{S}^{n-1})$. $\square$

Remark 4.8 (The Degenerate Morse-Bott Regime). If A has repeated eigenvalues, say $\lambda_k = \cdots = \lambda_{k+m}$, the critical set along that eigenspace forms a continuous critical submanifold $\mathbb{S}^m$ rather than isolated points. In this setting, f_A is a Morse-Bott function. The hypothesis of distinct eigenvalues in Theorem 4.7 is therefore strictly load-bearing.

4.4. Asymptotic Graphons and the Cut-Norm Topology ($n \rightarrow \infty$). In the functional domain, both W_A and W_B reside in the universal Lebesgue space $\mathcal{W} := \{W \in L^\infty([0, 1]^2) : W(x, y) = W(y, x)\}$.

Definition 4.9 (The Cut Norm (Frieze–Kannan / Lovász–Szegedy)). For any kernel $W \in L^1([0, 1]^2)$, the *Cut Norm* is defined by:

$$\|W\|_\square := \sup_{S, T \subseteq [0, 1]} \left| \int_{S \times T} W(x, y) dx dy \right|, \quad (4.29)$$

where the supremum is taken over all measurable subsets $S, T \subseteq [0, 1]$.

Theorem 4.10 (Duality with $L^\infty \rightarrow L^1$ Operator Norm and Grothendieck Inequality). *The cut norm $\|W\|_\square$ is equivalent to the operator norm of the integral operator $T_W: L^\infty([0, 1]) \rightarrow L^1([0, 1])$:*

$$\|W\|_\square \leq \|T_W\|_{L^\infty \rightarrow L^1} \leq 4\|W\|_\square, \quad (4.30)$$

where $\|T_W\|_{L^\infty \rightarrow L^1} := \sup_{\|u\|_\infty \leq 1, \|v\|_\infty \leq 1} \left| \iint_{[0,1]^2} W(x, y) u(x) v(y) dx dy \right|$.

Proof. For any indicator functions $u = \mathbf{1}_S$ and $v = \mathbf{1}_T$, we have $\langle u, T_W v \rangle = \int_{S \times T} W(x, y) dx dy$. Decomposing any u, v with $\|u\|_\infty \leq 1, \|v\|_\infty \leq 1$ into positive and negative parts $u = u_+ - u_-$ and $v = v_+ - v_-$:

$$\begin{aligned} \left| \int_{[0,1]^2} W uv dx dy \right| &\leq \left| \int W u_+ v_+ \right| + \left| \int W u_+ v_- \right| + \left| \int W u_- v_+ \right| + \left| \int W u_- v_- \right| \\ &\leq 4\|W\|_\square. \end{aligned} \quad (4.31)$$

Taking the supremum establishes the upper bound. The lower bound $\|W\|_\square \leq \|T_W\|_{L^\infty \rightarrow L^1}$ follows immediately by restricting $u = \mathbf{1}_S$ and $v = \mathbf{1}_T$. $\square$

Theorem 4.11 (Compactness of the Continuum Moduli Space). *Let $\bar{\mathcal{W}}$ be the space of symmetric graphons $W: [0, 1]^2 \rightarrow [0, 1]$, and let $\bar{\Pi}$ be the group of measure-preserving bijections. The quotient metric space $(\bar{\mathcal{W}}, \delta_\square) := (\bar{\mathcal{W}}/\bar{\Pi}, \delta_\square)$ is compact. Consequently, every sequence of symmetric matrices $\{A_n\}_{n=1}^\infty$ with entries uniformly bounded in a compact interval $[m, M]$ (under standard affine normalization $a_{ij} \mapsto \frac{a_{ij}-m}{M-m} \in [0, 1]$) has a subsequence whose step realizations $W_{A_{n_k}}$ converge in the cut metric to a limit graphon $W \in \bar{\mathcal{W}}$.*

Proof. See Lovász and Szegedy [8]. The compactness is a functional-analytic manifestation of the Szemerédi Regularity Lemma. $\square$

Example 4.12 (The Identity Matrix Continuum Collapse). Consider the identity matrix $I_n \in \mathbb{R}^{n \times n}$. In linear algebra, $\text{rank}(I_n) = n \rightarrow \infty$. However, its functional realization $W_{I_n}(x, y)$ is supported on n diagonal squares of area $1/n^2$, giving total support measure $n \times (1/n^2) = 1/n$. For any measurable sets $S, T \subseteq [0, 1]$:

$$\|W_{I_n}\|_\square \leq \iint_{[0,1]^2} W_{I_n}(x, y) dx dy = \frac{1}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (4.32)$$

Thus, in the continuous cut metric, $I_n \rightarrow 0$. Discrete full rank vanishes into the null graphon because the continuous diagonal has Lebesgue measure zero.

5. HIGH-DIMENSIONAL TENSORS: FUNCTIONAL GEOMETRY, CURVATURE, AND SCALED INVARIANTS

When transitioning from 2-tensors (matrices) to higher-order tensors $\mathcal{T} \in \bigotimes_{\alpha=1}^k \mathbb{R}^{d_\alpha}$ ($k \geq 3$), classical multilinear algebra confronts insurmountable algebraic barriers: tensor rank computation is NP-hard (Hillar and Lim [10]), and low-rank tensor varieties fail to be topologically closed. In this section, we develop the functional machinery that overcomes these bottlenecks and extends our theory to high-order continuum limits.

5.1. Variational Realization and Resolution of the Border-Rank Pathology. Let $\mathcal{T} \in \bigotimes_{\alpha=1}^k \mathbb{R}^{d_\alpha}$ be a k -th order tensor. The canonical CP approximation seeks:

$$\inf_{\mathcal{X} \in \mathcal{S}_r} \|\mathcal{T} - \mathcal{X}\|_F, \quad \text{where } \mathcal{S}_r := \left\{ \sum_{i=1}^r \lambda_i u_i^{(1)} \otimes \cdots \otimes u_i^{(k)} \right\}. \quad (5.1)$$

Theorem 5.1 (Functional Well-Posedness over Compact Product Manifolds). *Define the product of unit spheres $\mathcal{M} := \prod_{\alpha=1}^k \mathbb{S}^{d_\alpha-1}$, which is compact under the product Riemannian metric. The functional realization:*

$$\Phi_{\text{multi}}(\mathcal{T})(u^{(1)}, \dots, u^{(k)}) := \sum_{j_1, \dots, j_k} \mathcal{T}_{j_1 \dots j_k} u_{j_1}^{(1)} \dots u_{j_k}^{(k)} \quad (5.2)$$

is smooth: $\Phi_{\text{multi}}(\mathcal{T}) \in C^\infty(\mathcal{M})$. Consequently:

(i) The maximal multilinear singular value:

$$\sigma_{\max}(\mathcal{T}) := \sup_{(u^{(1)}, \dots, u^{(k)}) \in \mathcal{M}} \Phi_{\text{multi}}(\mathcal{T})(u^{(1)}, \dots, u^{(k)}) \quad (5.3)$$

is strictly attained at an extremal vector tuple $(u_*^{(1)}, \dots, u_*^{(k)}) \in \mathcal{M}$.

(ii) The critical points on $\mathcal{M}$ satisfy the Euler–Lagrange equations:

$$\mathcal{T}(u^{(1)}, \dots, u^{(\alpha-1)}, \cdot, u^{(\alpha+1)}, \dots, u^{(k)}) = \sigma u^{(\alpha)}, \quad \forall \alpha \in \{1, \dots, k\}, \quad (5.4)$$

which unconditionally recover the Lim–Qi singular value equations [10, 12] without diverging factor weights.

Proof. Since each sphere $\mathbb{S}^{d_\alpha-1}$ is compact, by Tychonoff’s Theorem $\mathcal{M}$ is compact. The polynomial mapping $\Phi_{\text{multi}}(\mathcal{T})$ is continuous on $\mathcal{M}$. By the Weierstrass Extreme Value Theorem, $\Phi_{\text{multi}}(\mathcal{T})$ attains its supremum and infimum on $\mathcal{M}$.

To establish (ii), form the Lagrangian on $\prod \mathbb{R}^{d_\alpha} \times \mathbb{R}^k$:

$$\mathcal{L}(u^{(1)}, \dots, u^{(k)}, \mu_1, \dots, \mu_k) = \Phi_{\text{multi}}(\mathcal{T})(u^{(1)}, \dots, u^{(k)}) - \frac{1}{2} \sum_{\alpha=1}^k \mu_\alpha (\|u^{(\alpha)}\|_2^2 - 1). \quad (5.5)$$

Vanishing of the gradient with respect to $u^{(\alpha)}$ gives:

$$\nabla_{u^{(\alpha)}} \Phi = \mu_\alpha u^{(\alpha)}. \quad (5.6)$$

Taking inner products with $u^{(\alpha)}$ and using multilinearity:

$$\langle u^{(\alpha)}, \nabla_{u^{(\alpha)}} \Phi \rangle = \Phi_{\text{multi}}(\mathcal{T}) = \mu_\alpha \|u^{(\alpha)}\|_2^2 = \mu_\alpha. \quad (5.7)$$

Since $\|u^{(\alpha)}\|_2 = 1$ for all α , we must have $\mu_1 = \dots = \mu_k = \sigma$. This proves that all singular equations share an identical eigenvalue parameter σ , eliminating the arbitrary scale divergences that afflict discrete CP-ALS iterations. $\square$

5.2. Continuum Limits of Dense Hyper-Networks: Hypergraphons. Just as symmetric matrices represent graphs, order- k tensors $\mathcal{T} \in \bigotimes^k \mathbb{R}^n$ represent k -uniform hypergraphs (e.g., k -way multi-body interactions).

Definition 5.2 (k -Hypergraphon and Multilinear Cut Norm). A k -hypergraphon is a symmetric measurable function $W: [0, 1]^k \rightarrow [0, 1]$. The *Multilinear Cut Norm* of order k is defined by:

$$\|W\|_{\square, k} := \sup_{S_1, \dots, S_k \subseteq [0, 1]} \left| \int_{S_1 \times \dots \times S_k} W(x_1, \dots, x_k) dx_1 \dots dx_k \right|, \quad (5.8)$$

where the supremum is taken over all measurable subsets $S_1, \dots, S_k \subseteq [0, 1]$.

Theorem 5.3 (Multilinear Operator Duality and Hypergraphon Compactness). *Let $\mathcal{W}_k$ be the space of k -hypergraphons endowed with the cut distance:*

$$\delta_{\square, k}(U, W) := \inf_{\psi \in \bar{\Pi}} \|U - W^\psi\|_{\square, k}, \quad (5.9)$$

where $\bar{\Pi}$ is the group of measure-preserving bijections of $[0, 1]$. Then:

(i) The cut norm is equivalent to the k -multilinear operator norm from $L^\infty \rightarrow L^1$:

$$\|W\|_{\square, k} \leq \|T_W\|_{L^\infty \times \dots \times L^\infty \rightarrow \mathbb{R}} \leq 2^k \|W\|_{\square, k}. \quad (5.10)$$

(ii) The quotient moduli space $(\widetilde{\mathcal{W}}_k, \delta_{\square, k})$ is compact. Every sequence of high-order tensors $\{\mathcal{T}_n\}_{n=1}^\infty$ with bounded entries possesses a convergent subsequence in the hypergraphon cut metric as $n \rightarrow \infty$.

Proof. The proof generalizes Theorem 4.10. Decomposing each test function $u^{(\alpha)} \in L^\infty([0, 1])$ with $\|u^{(\alpha)}\|_\infty \leq 1$ into positive and negative parts $u_+^{(\alpha)} - u_-^{(\alpha)}$, the multilinear integral splits into 2^k signed orthant integrals, each bounded by $\|W\|_{\square, k}$. The compactness of $\widetilde{\mathcal{W}}_k$ follows from the hypergraph regularity lemma of Gowers [7] (see also Lovász [9]). $\square$

5.3. Multivariate Sobolev Regularization on Transformer Attention Tensors. In modern deep learning architectures (Transformers), self-attention computes an activation tensor $\mathcal{A} \in \mathbb{R}^{B \times H \times N \times N}$, where B is batch size, H is number of attention heads, and N is sequence length:

$$\mathcal{A}_{b,h,i,j} = \text{Softmax} \left(\frac{Q_{b,h,i,:} K_{b,h,j,:}^T}{\sqrt{d_k}} \right). \quad (5.11)$$

Theorem 5.4 (Dirichlet Stability of Attention Fields). *Let $f_{\mathcal{A}_{b,h}}(x, y) \in H^2(\mathbb{T}^2)$ be the harmonic realization of the attention slice $\mathcal{A}_{b,h,:}$. If the network training loss includes the multilinear Sobolev penalty controlling the full $H^2(\mathbb{T}^2)$ norm:*

$$\begin{aligned} \mathcal{R}_{\text{Attn}}(\mathcal{A}) &:= \sum_{b=1}^B \sum_{h=1}^H \iint_{\mathbb{T}^2} (|f_{\mathcal{A}_{b,h}}|^2 + \|\nabla f_{\mathcal{A}_{b,h}}\|_2^2 + \lambda |\Delta f_{\mathcal{A}_{b,h}}|^2) \, dx \, dy \\ &= \sum_{b=1}^B \sum_{h=1}^H \|f_{\mathcal{A}_{b,h}}\|_{H^2(\mathbb{T}^2)}^2, \end{aligned} \quad (5.12)$$

then the attention maps satisfy a uniform Sobolev embedding $H^2(\mathbb{T}^2) \hookrightarrow C^{0,\alpha}(\mathbb{T}^2)$ ($\alpha \in (0, 1)$). Consequently, isolated adversarial attention spikes are bounded by:

$$\sup_{(x,y) \in \mathbb{T}^2} |f_{\mathcal{A}_{b,h}}(x, y)| \leq C_{\text{Sobolev}} \sqrt{\mathcal{R}_{\text{Attn}}(\mathcal{A})}, \quad (5.13)$$

which mathematically suppresses attention hallucination across ultra-long token sequences ($N \rightarrow \infty$).

Proof. In dimension 2, the Sobolev embedding theorem guarantees that the Sobolev space $H^2(\mathbb{T}^2)$ embeds continuously into the Hölder space $C^{0,\alpha}(\mathbb{T}^2)$ for all $\alpha \in (0, 1)$. By Morrey's inequality and elliptic Sobolev theory:

$$\|f\|_{L^\infty(\mathbb{T}^2)} \leq C_{\text{Sobolev}} \|f\|_{H^2(\mathbb{T}^2)} = C_{\text{Sobolev}} (\|f\|_{L^2}^2 + \|\nabla f\|_{L^2}^2 + \lambda \|\Delta f\|_{L^2}^2)^{1/2}. \quad (5.14)$$

Furthermore, because each row of the discrete attention matrix $\mathcal{A}_{b,h,:}$ is normalized via softmax ($\sum_{j=1}^N \mathcal{A}_{b,h,i,j} = 1$ with entries $\mathcal{A}_{b,h,i,j} \in [0, 1]$), the L^2 -norm $\|f_{\mathcal{A}}\|_{L^2} = \|\mathcal{A}\|_F \leq \sqrt{N}$ is bounded by the stochastic nature of the attention weights. Combining the full H^2 penalty with the Sobolev embedding constant C_{Sobolev} yields the claimed bound on the supremum. $\square$

5.4. Complexity of the High-Dimensional Tensor Morse Landscape.

Theorem 5.5 (Complexity of the Tensor Morse Landscape). *Let $\mathcal{T} \in \text{Sym}^d(\mathbb{R}^n)$ be a random Gaussian symmetric tensor whose entries are independent centered Gaussian variables with variance $\mathbb{E}[\mathcal{T}_{i_1 \dots i_d}^2] = \frac{1}{n^{d-1}}$. Then the expected number of critical points of the spherical realization $f_{\mathcal{T}}: \mathbb{S}^{n-1} \rightarrow \mathbb{R}$ grows exponentially with dimension:*

$$\mathbb{E}[|\text{Crit}(f_{\mathcal{T}})|] \sim C(d) \exp(n \Theta(d)) \quad \text{as } n \rightarrow \infty. \quad (5.15)$$

Proof. By the Kac-Rice formula for random fields on Riemannian manifolds (see Adler and Taylor [1]), the expected number of critical points is given by:

$$\mathbb{E}[|\text{Crit}(f_{\mathcal{T}})|] = \int_{\mathbb{S}^{n-1}} \mathbb{E} [|\det(\nabla^2 f_{\mathcal{T}}(x))| \mid \nabla f_{\mathcal{T}}(x) = 0] p_{\nabla f_{\mathcal{T}}(x)}(0) d\sigma(x), \quad (5.16)$$

where $\nabla^2 f_{\mathcal{T}}(x)$ is the covariant Riemannian Hessian on $T_x \mathbb{S}^{n-1}$ and $p_{\nabla f_{\mathcal{T}}(x)}(0)$ is the marginal density of the gradient at zero. Under the spherical symmetry of the covariance kernel $K(x, y) = \frac{1}{d}(x \cdot y)^d$, the field $f_{\mathcal{T}}$ and its gradient are isotropic. Conditional on $\nabla f_{\mathcal{T}}(x) = 0$, the covariant Hessian splits into:

$$\nabla^2 f_{\mathcal{T}}(x) = H_0 - (d-1)f_{\mathcal{T}}(x)I_{n-1}, \quad (5.17)$$

where H_0 is an $(n-1) \times (n-1)$ Gaussian Orthogonal Ensemble (GOE) matrix independent of $f_{\mathcal{T}}(x)$. By Wigner's semicircle law and the large deviations principle for the empirical spectral measure of GOE matrices, the expected absolute determinant scales asymptotically as $\exp(n\Sigma(d))$ where $\Sigma(d) = \frac{1}{2} \log(d-1) - \frac{d-2}{2(d-1)} > 0$ for $d \geq 3$ (Auffinger, Ben Arous, and Cérony [3], Theorem 1.1). Integrating across the sphere of volume $\text{Vol}(\mathbb{S}^{n-1})$ and over the energy levels yields the exponential growth $\sim C(d) \exp(n\Theta(d))$. $\square$

5.5. Critical Scaling and Measure Concentration in Infinite Dimensions ($d \rightarrow \infty$).

A central dilemma in high-dimensional tensor analysis is establishing the exact normalization that prevents multilinear potentials from either exploding or vanishing as the mode dimension $d \rightarrow \infty$.

Theorem 5.6 (Critical Multilinear Scaling and Sub-Gaussian Concentration). *Let $\mathcal{T} \in (\mathbb{R}^d)^{\otimes k}$ be a random Gaussian tensor whose entries are independent standard normal variables: $\mathcal{T}_{j_1 \dots j_k} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Define the normalized functional realization on unit spheres:*

$$\hat{f}_{\mathcal{T}}(u^{(1)}, \dots, u^{(k)}) := \frac{1}{\sqrt{d}} \sum_{j_1, \dots, j_k=1}^d \mathcal{T}_{j_1 \dots j_k} u_{j_1}^{(1)} \dots u_{j_k}^{(k)}, \quad u^{(\alpha)} \in \mathbb{S}^{d-1}. \quad (5.18)$$

Then:

- (i) The normalization exponent $\alpha = 1/2$ is the unique critical scaling ensuring that the operator norm over unit spheres remains bounded and non-trivial:

$$\mathbb{E} \left[\sup_{u^{(\alpha)} \in \mathbb{S}^{d-1}} |\hat{f}_{\mathcal{T}}(u^{(1)}, \dots, u^{(k)})| \right] = \Theta(1) \quad \text{as } d \rightarrow \infty. \quad (5.19)$$

- (ii) For any independent tuple of random vectors $(u^{(1)}, \dots, u^{(k)})$ drawn uniformly from the Haar measure on $\mathbb{S}^{d-1}$, the functional satisfies the sharp sub-Gaussian concentration inequality:

$$\mathbb{P} \left(|\hat{f}_{\mathcal{T}}(u^{(1)}, \dots, u^{(k)})| > \epsilon \right) \leq 2 \exp(-c_k d \epsilon^2), \quad (5.20)$$

where $c_k > 0$ is a universal dimensional constant depending only on k .

Remark 5.7 (Spin-Glass Extensive Scaling vs Geometric Operator Norm). In the statistical physics of spherical p -spin glasses ($p = k$), spin vectors are traditionally normalized to length $\|\sigma^{(\alpha)}\|_2 = \sqrt{d}$ so that each spin coordinate has variance 1. In that convention, the multilinear sum pulls out a factor of $(\sqrt{d})^k = d^{k/2}$. The Hamiltonian is normalized by $d^{-(k-1)/2}$, yielding an extensive Hamiltonian $H_k(\sigma) = d^{-(k-1)/2} \sum \mathcal{T}_{j_1 \dots j_k} \sigma_{j_1}^{(1)} \dots \sigma_{j_k}^{(k)} = \sqrt{d} \sum \mathcal{T} u^{(1)} \dots u^{(k)}$. The intensive energy per particle $\frac{1}{d} H_k(\sigma) = \frac{1}{\sqrt{d}} \sum \mathcal{T} u^{(1)} \dots u^{(k)} = \hat{f}_{\mathcal{T}}(u)$ is of order $\Theta(1)$, reconciling the physical Hamiltonian exponent $(k-1)/2$ with the geometric unit-sphere normalization factor $1/\sqrt{d}$.

Proof. To prove (i), let $\mathcal{M} = \prod_{\alpha=1}^k \mathbb{S}^{d-1}$ be the product manifold of unit spheres. For an arbitrary fixed tuple $v \in \mathcal{M}$, the multilinear evaluation:

$$X(v) := \sum_{j_1, \dots, j_k=1}^d \mathcal{T}_{j_1 \dots j_k} v_{j_1}^{(1)} \dots v_{j_k}^{(k)} \quad (5.21)$$

is an exact linear combination of independent standard Gaussian random variables with variance:

$$\text{Var}(X(v)) = \sum_{j_1, \dots, j_k=1}^d (v_{j_1}^{(1)})^2 \dots (v_{j_k}^{(k)})^2 = \prod_{\alpha=1}^k \|v^{(\alpha)}\|_2^2 = 1. \quad (5.22)$$

Hence $X(v) \sim \mathcal{N}(0, 1)$ is identically a standard normal scalar for every $v \in \mathcal{M}$. Let $\mathcal{N}_\epsilon$ be an optimal ϵ -net of $\mathbb{S}^{d-1}$ with cardinality $|\mathcal{N}_\epsilon| \leq (1 + 2/\epsilon)^d$. The product net $\mathcal{N} := \prod_{\alpha=1}^k \mathcal{N}_\epsilon \subset \mathcal{M}$ satisfies $|\mathcal{N}| \leq (1 + 2/\epsilon)^{kd}$. By the standard Gaussian maximal inequality over the finite net $\mathcal{N}$:

$$\mathbb{E} \left[\sup_{v \in \mathcal{N}} |X(v)| \right] \leq \sqrt{2 \log(2|\mathcal{N}|)} \leq \sqrt{2kd \log(1 + 2/\epsilon) + 2 \log 2} = C(k, \epsilon) \sqrt{d}. \quad (5.23)$$

By multilinear approximation on $\mathcal{M}$, choosing $\epsilon = \frac{1}{2k}$ guarantees that $\sup_{u \in \mathcal{M}} |X(u)| \leq \frac{1}{1-k\epsilon} \sup_{v \in \mathcal{N}} |X(v)| \leq 2 \sup_{v \in \mathcal{N}} |X(v)|$. Therefore:

$$\mathbb{E} \left[\sup_{u \in \mathcal{M}} |X(u)| \right] \leq 2C(k) \sqrt{d} = \mathcal{O}(\sqrt{d}). \quad (5.24)$$

Conversely, restricting $u^{(1)} = \dots = u^{(k-1)} = e_1$ yields the Gaussian contraction $\sum_{j_k} \mathcal{T}_{1 \dots 1 j_k} u_{j_k}^{(k)}$. Its supremum over $u^{(k)} \in \mathbb{S}^{d-1}$ is the Euclidean norm of the standard Gaussian vector $Z := (\mathcal{T}_{1 \dots 1 1}, \dots, \mathcal{T}_{1 \dots 1 d})^T \sim \mathcal{N}(0, I_d)$, which satisfies $\mathbb{E}[\|Z\|_2] = \sqrt{2} \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} = \sqrt{d}(1 - \mathcal{O}(1/d)) = \Omega(\sqrt{d})$. Thus the unnormalized expected supremum is strictly $\Theta(\sqrt{d})$. Dividing by $\sqrt{d} = d^{1/2}$ yields $\mathbb{E}[\sup |\hat{f}_{\mathcal{T}}|] = \Theta(1)$, establishing uniqueness of the scaling exponent 1/2 on unit spheres.

To establish (ii), recall that the round sphere $\mathbb{S}^{d-1}$ has Ricci curvature $\text{Ric}(\mathbb{S}^{d-1}) = (d-2)g_{\mathbb{S}}$. By the Lévy–Gromov isoperimetric theorem and the Bakry–Émery criterion, any Lipschitz function $F: \mathbb{S}^{d-1} \rightarrow \mathbb{R}$ with Lipschitz constant $\|F\|_{\text{Lip}} \leq L$ satisfies:

$$\mathbb{P}(|F - \mathbb{E}[F]| > \epsilon) \leq 2 \exp \left(-\frac{(d-1)\epsilon^2}{2L^2} \right). \quad (5.25)$$

Conditioning on $(u^{(2)}, \dots, u^{(k)})$, the function $u^{(1)} \mapsto \hat{f}_{\mathcal{T}}(u^{(1)}, \dots, u^{(k)})$ is linear on $\mathbb{S}^{d-1}$ with Lipschitz constant $L_1 = \frac{1}{\sqrt{d}} \|w\|_2$, where $w_i := \sum_{j_2, \dots, j_k} \mathcal{T}_{i, j_2 \dots j_k} u_{j_2}^{(2)} \dots u_{j_k}^{(k)}$. Since $\mathcal{T}$ is Gaussian, the coordinates $w_1, \dots, w_d$ are independent centered Gaussian variables with variance $\prod_{\alpha=2}^k \|u^{(\alpha)}\|_2^2 = 1$. Hence $w \sim \mathcal{N}(0, I_d)$ and $\|w\|_2^2 \sim \chi^2(d)$. By the Laurent–Massart tail bound for chi-squared distributions, $\mathbb{P}(\|w\|_2^2 \geq 2d) \leq \exp(-d/8)$. On the typical event $\{\|w\|_2^2 \leq 2d\}$, the Lipschitz constant satisfies $L_1 \leq \sqrt{2}$. Applying the Herbst argument inductively across all k spheres and unconditioning via the exponential chi-squared tail yields (5.20). $\square$

6. DEEP IMPLICATIONS ACROSS THE MATHEMATICAL SCIENCES

The unification of discrete multilinear algebra with infinite-dimensional functional analysis provides rigorous solutions to long-standing structural impasses across five major scientific domains.

6.1. Deep Learning: Sobolev Weight Regularization and Zero-Shot Width Scaling.

In standard deep neural network training, weights $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ are conventionally regularized via coordinate-blind penalties: Weight Decay $\|W\|_F^2$ or Spectral Norm constraints $\sigma_{\max}(W) \leq 1$. By Theorem 4.1, two weight matrices with identical spectral profiles can exhibit a factor of $\Theta(n^2)$ difference in their Dirichlet spatial energy $\mathcal{E}(f_W)$.

This spectral blindness leaves overparametrized networks susceptible to high-frequency adversarial perturbations. We propose the *Sobolev Regularized Loss*:

$$\mathcal{L}_{\text{Sobolev}}(\theta) := \mathcal{L}_{\text{task}}(\theta) + \sum_{\ell=1}^L \lambda_{\ell} \|\nabla f_{W_{\ell}}\|_{L^2(\mathbb{T}^2)}^2. \quad (6.1)$$

Penalizing the functional Dirichlet energy forces weight tensors into lower-frequency Sobolev bands $H^1(\mathbb{T}^2)$, guaranteeing Lipschitz stability of feature maps without constraining expressive capacity.

Furthermore, by Theorem 4.11, dense weight matrices of growing dimensions $W_n \in \mathbb{R}^{n \times n}$ converge in the cut metric $\delta_{\square}$ to a continuous limit graphon $\mathcal{W}_{\infty} \in \mathcal{W}$. This provides a mathematically sound protocol for *Zero-Shot Width Scaling*: a network trained at width $n = 10^3$ can be transferred to an ultra-large architecture at $n = 10^6$ parameters by sampling the continuous kernel $\mathcal{W}_{\infty}(\frac{i}{n}, \frac{j}{n})$, completely eliminating the computational cost of training mega-models from scratch.

6.2. Theoretical Physics: Spin Glass Transitions and Parisi Symmetry Breaking.

In the statistical physics of disordered systems, the p -spin spherical spin glass is defined by the Hamiltonian $H_n(\sigma) = \sum_{i_1, \dots, i_p} J_{i_1 \dots i_p} \sigma_{i_1} \dots \sigma_{i_p}$ on the spin sphere $\sum \sigma_i^2 = n$. Determining the ground state energy and partition function $Z = \int e^{-\beta H_n} d\sigma$ via discrete tensor algebra is NP-hard.

Our functional realization on $\mathbb{S}^{n-1}$ (Theorem 5.1 and 5.3) proves that the physics of spin glasses is fundamentally governed by the *Morse landscape topology*. The exponential explosion of saddles $|\text{Crit}(f_J)| \sim \exp(n\Theta(p))$ proves that below a critical temperature T_c , the sub-level sets:

$$M_E := \{\sigma \in \mathbb{S}^{n-1} : f_J(\sigma) \leq E\} \quad (6.2)$$

undergo massive homological shattering into disconnected topological pockets. Giorgio Parisi's celebrated Replica Symmetry Breaking (RSB) corresponds precisely to the onset of non-trivial zeroth Betti numbers $\beta_0(M_E) \gg 1$ in the functional sub-level sets, establishing a rigorous topological foundation for glassy phase transitions.

6.3. Complex Networks: Universal Scale-Free Metrics on Massive Graphs.

When comparing real-world complex networks of disparate dimensions (e.g., biological connectomes or internet topologies across developmental epochs, where $|V_1| = 10^5$ and $|V_2| = 10^8$), discrete matrix algebra is paralyzed: the difference $A_1 - A_2$ is undefined because the underlying vector spaces are non-isomorphic.

Under our step realization Φ_{step} , every adjacency matrix A becomes a function $W_A : [0, 1]^2 \rightarrow [0, 1]$. Theorem 4.10 establishes that the cut distance $\delta_{\square}(W_{A_1}, W_{A_2})$ embeds networks of *arbitrary, unequal sizes* into the universal compact metric space $(\widetilde{\mathcal{W}}, \delta_{\square})$. This unlocks:

- (i) Rigorous definition of time derivatives $\frac{d}{dt} W(t)$ for networks whose node counts continuously grow;
- (ii) Statistical hypothesis testing comparing graph families across different sampling scales;
- (iii) Continuous epidemic and consensus dynamics modeled via integro-differential equations rather than discrete master equations.

6.4. Computational Multilinear Algebra: Circumventing the De Silva–Lim Pathology. In multilinear data processing (signal processing, hyper-spectral imaging, quantum chemistry), computing the best low-rank tensor approximation under the CP format:

$$\min_{\text{rank}(\mathcal{X}) \leq r} \|\mathcal{T} - \mathcal{X}\|_F \quad (6.3)$$

routinely encounters fatal numerical instabilities. In their landmark work, De Silva and Lim [4, 10] proved that for tensors of order $k \geq 3$, the variety $\mathcal{S}_r := \{\mathcal{X} : \text{rank}(\mathcal{X}) \leq r\}$ is *not topologically closed* in the Euclidean topology. Consequently, standard algorithms such as Alternating Least Squares (CP-ALS) diverge, with factor weights diverging to $\pm\infty$.

Our functional framework circumvents this pathology: by realizing the tensor as a multilinear form $\Phi(\mathcal{T})$ on the compact manifold $\mathcal{M} = \mathbb{S}^{n_1-1} \times \dots \times \mathbb{S}^{n_k-1}$, the optimization problem transforms into finding critical points of a smooth potential on a compact domain. By the Weierstrass Extreme Value Theorem, the functional extrema are *unconditionally guaranteed to exist*. The variational singular vectors (the Lim–Qi equations [10, 12] (5.4)) are well-posed, yielding a robust framework for tensor factorizations that never diverges.

6.5. Biomedical Imaging: Hausdorff Perimeters and Minimal Surface Level Curves.

In medical tomography (MRI, CT) and computer vision, digital images are discrete matrices of pixel intensities $A \in \mathbb{R}^{M \times N}$. The standard Frobenius norm $\|A\|_F^2 = \sum a_{ij}^2$ penalizes image energy diffusively, failing to distinguish between smooth background gradients and sharp organ boundaries.

By Theorem 4.6, the Total Variation $\text{TV}(W_A)$ of the matrix step realization is mathematically identical to the *integrated 1-dimensional Hausdorff measure* (arc-length) of the essential level curves:

$$\text{TV}(W_A) = \int_{-\infty}^{\infty} \mathcal{H}^1(\partial^* \{W_A > t\}) dt. \quad (6.4)$$

This establishes that discrete Total Variation regularization (such as the Rudin–Osher–Fatemi model [13]) is not merely a heuristic filter, but a precise variational minimal-surface formulation. Matrices with identical pixel histograms possess wildly divergent level-set perimeters, providing a quantitative metric for measuring the fractal geometric boundary complexity of pathological tissue lesions directly from raw array data.

7. SYNTHESIS AND COMPARISON TABLE

Table 1 synthesizes the correspondence between classical algebraic matrix invariants and their emerging functional counterparts.

Entity	Algebraic Invariant	Functional Realization	Emerging Invariant / Impact
$A \in \mathbb{R}^{n \times n}$	Spectrum $\sigma(A)$, $\ A\ _F$	$f_A \in H^s(\mathbb{T}^2)$	Dirichlet Energy $\mathcal{E}(f_A)$, Sobolev DL regularization.
$A \in \mathbb{R}^{n \times n}$	Finite differences ΔA	$W_A \in \text{BV}((0, 1)^2)$	Total Variation TV, Level-set perimeter $\mathcal{H}^1(\partial^* E_t)$.
$A \in \text{Sym}_n$	Spectrum $\lambda_1 \leq \dots \leq \lambda_n$	$f_A \in C^\infty(\mathbb{S}^{n-1})$	Morse indices $\gamma(v_i) = i - 1$, Euler characteristic χ .
$A (n \rightarrow \infty)$	Rank, spectral radius ρ	Graphon $W_A \in \widetilde{\mathcal{W}}$	Cut norm $\ W_A\ _\square$, Universal scale-free network metric.
$\mathcal{T} \in \text{Sym}^d$	Tensor rank, Z-eigenvalues	$f_{\mathcal{T}} \in C^\infty(\mathbb{S}^{n-1})$	Spin glass Morse complexity, De Silva–Lim resolution.

TABLE 1. Systematic bridge between discrete multilinear algebra and emerging functional-geometric invariants.

8. CONCLUSION AND OPEN PROBLEMS

We have formulated a comprehensive theoretical framework establishing that matrices and higher-order tensors are not merely discrete algebraic arrays, but discrete projections of rich continuous functional landscapes. Through Functional Realization Operators Φ , we showed that properties traditionally considered external or non-algebraic—such as spatial smoothness, total variation, perimeter of level curves, Morse critical topologies, and infinite-dimensional graphon limits—arise naturally and rigorously.

Several fundamental open problems invite future research:

- (1) **Optimal Realization Bounds:** For a given matrix A , what is the infimum Dirichlet energy $\inf_{\pi \in S_n} \mathcal{E}(f_{P_\pi A P_\pi^T})$ over all coordinate permutations, and can this combinatorial-functional optimization problem be relaxed to a tractable Riemannian flow?
- (2) **Hypergraphon Curvature:** Can one define a Ricci curvature tensor directly on higher-order tensor realizations (hypergraphons) that governs the convergence of message-passing algorithms on dense hypergraphs?
- (3) **Topological Data Analysis (TDA) of Tensor Landscapes:** How do persistent homology barcodes of the sublevel sets $\{f_{\mathcal{T}} \leq c\}$ correlate with the numerical rank and tensor train (TT) decomposition ranks of $\mathcal{T}$?

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